```
base) Sergio@SERHIIs-MacBook-Pro homework_buggy % pytest -v tests/
test_iot_recommendations.py
========================================= test session starts
=========================================
platform darwin -- Python 3.9.12, pytest-8.4.2, pluggy-1.6.0 -- /Users/Sergio/
Documents/QA_Embedded/Gate/Task7/HomeWork/homework_buggy/.venv/bin/
python
cachedir: .pytest_cache
metadata: {'Python': '3.9.12', 'Platform': 'macOS-10.16-x86_64-i386-64bit',
'Packages': {'pytest': '8.4.2','pluggy': '1.6.0'}, 'Plugins': {'metadata': '3.1.1',
'html': '4.2.0'}, 'JAVA_HOME': '/Library/Java/JavaVirtualMachines/jdk-21.jdk/
Contents/Home'}
rootdir: /Users/Sergio/Documents/QA_Embedded/Gate/Task7/HomeWork/
homework_buggy
plugins: metadata-3.1.1, html-4.2.0
collected 27 items

tests/
test_iot_recommendations.py::test_rec1a_buffer_never_exceeds_capacity
FAILED                  [  3%]
tests/test_iot_recommendations.py::test_rec1b_dropped_counter_and_log
FAILED                    [  7%]
tests/test_iot_recommendations.py::test_rec1c_oldest_dropped_newest_kept
FAILED                    [ 11%]
tests/test_iot_recommendations.py::test_rec1d_stress_offline_100 FAILED
[ 14%]
tests/
test_iot_recommendations.py::test_rec2a_each_buffered_message_published
_once FAILED           [ 18%]
tests/test_iot_recommendations.py::test_rec2b_flush_is_fifo FAILED
[ 22%]
tests/
test_iot_recommendations.py::test_rec2c_buffer_empty_and_sent_correct_aft
er_flush FAILED       [ 25%]
tests/test_iot_recommendations.py::test_rec2d_no_phantom_ids PASSED
[ 29%]
tests/test_iot_recommendations.py::test_rec2e_flush_after_overflow FAILED
[ 33%]
tests/test_iot_recommendations.py::test_rec3a_no_leak_in_stress FAILED
[ 37%]
tests/
test_iot_recommendations.py::test_rec3b_no_leak_in_buffer_flush_cycles
FAILED                  [ 40%]
tests/test_iot_recommendations.py::test_rec4a_backoff_grows FAILED
[ 44%]
tests/test_iot_recommendations.py::test_rec4b_no_reconnect_storm FAILED
[ 48%]
```

```
tests/test_iot_recommendations.py::test_rec4c_interval_respects_backoff
FAILED                    [ 51%]
tests/test_iot_recommendations.py::test_rec4d_recovers_fast_after_net_up
PASSED                    [ 55%]
tests/
test_iot_recommendations.py::test_rec5a_sent_plus_dropped_equals_generat
ed FAILED             [ 59%]
tests/test_iot_recommendations.py::test_rec5b_sent_equals_publish_lines
PASSED                    [ 62%]
tests/
test_iot_recommendations.py::test_rec5c_duplicates_counter_matches_log
PASSED                [ 66%]
tests/test_iot_recommendations.py::test_rec5d_stress_counters_connected
PASSED                    [ 70%]
tests/test_iot_recommendations.py::test_rec6a_stress_invalid_rejected[-5]
FAILED                    [ 74%]
tests/test_iot_recommendations.py::test_rec6a_stress_invalid_rejected[1.5]
FAILED                    [ 77%]
tests/test_iot_recommendations.py::test_rec6a_stress_invalid_rejected[abc]
FAILED                    [ 81%]
tests/test_iot_recommendations.py::test_rec6a_stress_invalid_rejected[10 20]
FAILED                [ 85%]
tests/
test_iot_recommendations.py::test_rec6b_stress_out_of_range_reported[0]
FAILED                [ 88%]
tests/
test_iot_recommendations.py::test_rec6b_stress_out_of_range_reported[501]
FAILED              [ 92%]
tests/test_iot_recommendations.py::test_rec6c_net_down_idempotent FAILED
[ 96%]
tests/
test_iot_recommendations.py::test_rec6d_net_up_when_connected_is_silent
FAILED                [100%]

================================================= FAILURES
=================================================
_____________________________________
test_rec1a_buffer_never_exceeds_capacity
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec1a_buffer_never_exceeds_capacity(node):
        node.offline_send(12)
        st = node.status()
>       assert st["buffer"] <= CAP, f"STATUS buffer={st['buffer']}"
E       AssertionError: STATUS buffer=12
```

```
E       assert 12 <= 8

tests/test_iot_recommendations.py:91: AssertionError
_____________________________________
test_rec1b_dropped_counter_and_log
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec1b_dropped_counter_and_log(node):
        node.offline_send(12)
>       assert node.status()["dropped"] == 4
E       assert 0 == 4

tests/test_iot_recommendations.py:99: AssertionError
_____________________________________
test_rec1c_oldest_dropped_newest_kept
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec1c_oldest_dropped_newest_kept(node):
        ids = node.offline_send(12)
>       assert published(node.cmd("NET UP", 10)) == ids[-CAP:]
E       AssertionError: assert [33, 33, 34, 34, 35, 35, ...] == [29, 30, 31, 32, 33,
34, ...]
E
E         At index 0 diff: 33 != 29
E         Left contains 16 more items, first extra item: 29
E
E         Full diff:
E           [
E         +     33,...
E
E         ...Full output truncated (24 lines hidden), use '-vv' to show

tests/test_iot_recommendations.py:106: AssertionError
_____________________________________ test_rec1d_stress_offline_100
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec1d_stress_offline_100(node):
        node.cmd("NET DOWN", 1); node.cmd("STRESS 100", 8); st =
node.status()
>       assert st["buffer"] <= CAP and st["dropped"] == 100 - CAP, st
E       AssertionError: {'broker': False, 'buffer': 100, 'capacity': 8, 'dropped':
```

```
0, ...}
E       assert (100 <= 8)

tests/test_iot_recommendations.py:111: AssertionError
_____________________________________
test_rec2a_each_buffered_message_published_once
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec2a_each_buffered_message_published_once(node):
        ids = node.offline_send(5)
        pub = published(node.cmd("NET UP", 8))
>       assert sorted(pub) == sorted(ids), f"queued {ids}, published {pub}"
E       AssertionError: queued [137, 138, 139, 140, 141], published [137, 137, 138,
138, 139, 139, 140, 140, 141, 141]
E       assert [137, 137, 13...139, 139, ...] == [137, 138, 139, 140, 141]
E
E         At index 1 diff: 137 != 138
E         Left contains 5 more items, first extra item: 139
E
E         Full diff:
E           [
E               137,...
E
E         ...Full output truncated (10 lines hidden), use '-vv' to show

tests/test_iot_recommendations.py:118: AssertionError
_____________________________________________ test_rec2b_flush_is_fifo
_____________________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec2b_flush_is_fifo(node):
        ids = node.offline_send(8)
>       assert published(node.cmd("NET UP", 8)) == ids
E       AssertionError: assert [142, 142, 14...144, 144, ...] == [142, 143, 14...146,
147, ...]
E
E         At index 1 diff: 142 != 143
E         Left contains 8 more items, first extra item: 146
E
E         Full diff:
E           [
E               142,...
E
E         ...Full output truncated (16 lines hidden), use '-vv' to show
```

```
tests/test_iot_recommendations.py:123: AssertionError
_____________________________
test_rec2c_buffer_empty_and_sent_correct_after_flush
_____________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec2c_buffer_empty_and_sent_correct_after_flush(node):
        node.offline_send(8); node.cmd("NET UP", 8)
        st = node.status()
>       assert st["net"] == "CONNECTED" and st["buffer"] == 0 and st["sent"]
== 8, st
E       AssertionError: {'broker': True, 'buffer': 0, 'capacity': 8, 'dropped': 0, ...}
E       assert ('CONNECTED' == 'CONNECTED'
E
E           CONNECTED and 0 == 0 and 16 == 8)

tests/test_iot_recommendations.py:129: AssertionError
_____________________________________ test_rec2e_flush_after_overflow
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec2e_flush_after_overflow(node):
        ids = node.offline_send(12)
        pub = published(node.cmd("NET UP", 10))
>       assert pub == ids[-CAP:], f"expected {ids[-CAP:]}, got {pub}"
E       AssertionError: expected [172, 173, 174, 175, 176, 177, 178, 179], got [176,
176, 177, 177, 178, 178, 179, 179, 172, 172, 173, 173, 174, 174, 175, 175, 176, 176,
177, 177, 178, 178, 179, 179]
E       assert [176, 176, 17...178, 178, ...] == [172, 173, 17...176, 177, ...]
E
E         At index 0 diff: 176 != 172
E         Left contains 16 more items, first extra item: 172
E
E         Full diff:
E           [
E         +     176,...
E
E         ...Full output truncated (24 lines hidden), use '-vv' to show

tests/test_iot_recommendations.py:141: AssertionError
_____________________________________ test_rec3a_no_leak_in_stress
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>
```

```
    def test_rec3a_no_leak_in_stress(node):
        heaps = [node.status()["heap"]]
        for _ in range(5):
            node.cmd("STRESS 100", 6); heaps.append(node.status()["heap"])
        per_msg = [(a - b) / 100 for a, b in zip(heaps, heaps[1:])]
        print("bytes/msg:", per_msg)
>       assert max(per_msg) < 4, f"leak {per_msg} B/message"
E       AssertionError: leak [79.92, 80.12, 80.0, 80.0, 80.0] B/message
E       assert 80.12 < 4
E        +  where 80.12 = max([79.92, 80.12, 80.0, 80.0, 80.0])

tests/test_iot_recommendations.py:151: AssertionError
----------------------------------------- Captured stdout call
-----------------------------------------
bytes/msg: [79.92, 80.12, 80.0, 80.0, 80.0]
_____________________________________
test_rec3b_no_leak_in_buffer_flush_cycles
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec3b_no_leak_in_buffer_flush_cycles(node):
        h0 = node.status()["heap"]
        for _ in range(5):
            node.offline_send(8); node.cmd("NET UP", 6)
        h1 = node.status()["heap"]
        print("heap", h0, "->", h1)
>       assert h0 - h1 < 500, f"heap lost {h0 - h1} B in 5 buffer/flush cycles"
E       AssertionError: heap lost 3200 B in 5 buffer/flush cycles
E       assert (296280 - 293080) < 500

tests/test_iot_recommendations.py:160: AssertionError
----------------------------------------- Captured stdout call
-----------------------------------------
heap 296280 -> 293080
_____________________________________________ test_rec4a_backoff_grows
_____________________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec4a_backoff_grows(node):
        bo = [b for _, b in retries(node.cmd("NET DOWN", 20))]
        assert bo, "no retry lines"
>       assert bo == sorted(bo), f"backoff decreases: {bo[:10]}"
E       AssertionError: backoff decreases: [500, 50, 50, 50, 50, 50, 50, 50, 50,
50]
```

```
E       assert [500, 50, 50, 50, 50, 50, ...] == [50, 50, 50, 50, 50, 50, ...]
E
E         At index 0 diff: 500 != 50
E
E         Full diff:
E           [
E         -     50,
E         -     50,...
E
E         ...Full output truncated (780 lines hidden), use '-vv' to show

tests/test_iot_recommendations.py:167: AssertionError
_____________________________________________ test_rec4b_no_reconnect_storm
_____________________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec4b_no_reconnect_storm(node):
        n = len(retries(node.cmd("NET DOWN", 20)))
>       assert n <= 12, f"{n} retries in 20 s"
E       AssertionError: 391 retries in 20 s
E       assert 391 <= 12

tests/test_iot_recommendations.py:173: AssertionError
_____________________________________________ test_rec4c_interval_respects_backoff
_____________________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec4c_interval_respects_backoff(node):
        r = retries(node.cmd("NET DOWN", 15))
        assert len(r) >= 2
        for (t0, b0), (t1, _) in zip(r, r[1:]):
>           assert t1 - t0 >= 0.8 * b0 / 1000 - 0.03, f"gap {t1 - t0:.3f}s < backoff
{b0} ms"
E           AssertionError: gap 0.001s < backoff 50 ms
E           assert (1792948248.4949398 - 1792948248.493739) >= (((0.8 * 50) /
1000) - 0.03)

tests/test_iot_recommendations.py:180: AssertionError
_____________________________________
test_rec5a_sent_plus_dropped_equals_generated
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec5a_sent_plus_dropped_equals_generated(node):
```

```
        node.offline_send(12); node.cmd("NET UP", 10)
        st = node.status()
>       assert (st["sent"], st["dropped"]) == (8, 4), st
E       AssertionError: {'broker': True, 'buffer': 0, 'capacity': 8, 'dropped': 0, ...}
E       assert (24, 0) == (8, 4)
E
E         At index 0 diff: 24 != 8
E
E         Full diff:
E           (
E         -     8,
E         -     4,...
E
E         ...Full output truncated (4 lines hidden), use '-vv' to show

tests/test_iot_recommendations.py:195: AssertionError
_____________________________________ test_rec6a_stress_invalid_rejected[-5]
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>, arg =
'-5'

    @pytest.mark.parametrize("arg", ["-5", "1.5", "abc", "10 20"])
    def test_rec6a_stress_invalid_rejected(node, arg):
        out = node.cmd(f"STRESS {arg}", 3)
>       assert any("ERROR" in l for _, l in out), "no [CLI] ERROR"
E       AssertionError: no [CLI] ERROR
E       assert False
E        +  where False = any(<generator object
test_rec6a_stress_invalid_rejected.<locals>.<genexpr> at 0x7fea46a15ac0>)

tests/test_iot_recommendations.py:220: AssertionError
_____________________________________ test_rec6a_stress_invalid_rejected[1.5]
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>, arg =
'1.5'

    @pytest.mark.parametrize("arg", ["-5", "1.5", "abc", "10 20"])
    def test_rec6a_stress_invalid_rejected(node, arg):
        out = node.cmd(f"STRESS {arg}", 3)
>       assert any("ERROR" in l for _, l in out), "no [CLI] ERROR"
E       AssertionError: no [CLI] ERROR
E       assert False
E        +  where False = any(<generator object
test_rec6a_stress_invalid_rejected.<locals>.<genexpr> at 0x7fea46a159e0>)
```

```
tests/test_iot_recommendations.py:220: AssertionError
_____________________________________ test_rec6a_stress_invalid_rejected[abc]
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>, arg =
'abc'

    @pytest.mark.parametrize("arg", ["-5", "1.5", "abc", "10 20"])
    def test_rec6a_stress_invalid_rejected(node, arg):
        out = node.cmd(f"STRESS {arg}", 3)
>       assert any("ERROR" in l for _, l in out), "no [CLI] ERROR"
E       AssertionError: no [CLI] ERROR
E       assert False
E        +  where False = any(<generator object
test_rec6a_stress_invalid_rejected.<locals>.<genexpr> at 0x7fea46bb50b0>)

tests/test_iot_recommendations.py:220: AssertionError
_____________________________________ test_rec6a_stress_invalid_rejected[10
20] _____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>, arg = '10
20'

    @pytest.mark.parametrize("arg", ["-5", "1.5", "abc", "10 20"])
    def test_rec6a_stress_invalid_rejected(node, arg):
        out = node.cmd(f"STRESS {arg}", 3)
>       assert any("ERROR" in l for _, l in out), "no [CLI] ERROR"
E       AssertionError: no [CLI] ERROR
E       assert False
E        +  where False = any(<generator object
test_rec6a_stress_invalid_rejected.<locals>.<genexpr> at 0x7fea46bb53c0>)

tests/test_iot_recommendations.py:220: AssertionError
_____________________________________
test_rec6b_stress_out_of_range_reported[0]
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>, arg = '0'

    @pytest.mark.parametrize("arg", ["0", "501"])
    def test_rec6b_stress_out_of_range_reported(node, arg):
        out = node.cmd(f"STRESS {arg}", 3)
>       assert any("ERROR" in l for _, l in out), "silently clamped (assumption:
must report)"
E       AssertionError: silently clamped (assumption: must report)
E       assert False
```

```
E       +  where False = any(<generator object
test_rec6b_stress_out_of_range_reported.<locals>.<genexpr> at
0x7fea46bb5890>)

tests/test_iot_recommendations.py:227: AssertionError
_____________________________________
test_rec6b_stress_out_of_range_reported[501]
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>, arg =
'501'

    @pytest.mark.parametrize("arg", ["0", "501"])
    def test_rec6b_stress_out_of_range_reported(node, arg):
        out = node.cmd(f"STRESS {arg}", 3)
>       assert any("ERROR" in l for _, l in out), "silently clamped (assumption:
must report)"
E       AssertionError: silently clamped (assumption: must report)
E       assert False
E        +  where False = any(<generator object
test_rec6b_stress_out_of_range_reported.<locals>.<genexpr> at
0x7fea46bb5d60>)

tests/test_iot_recommendations.py:227: AssertionError
_____________________________________________ test_rec6c_net_down_idempotent
_____________________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec6c_net_down_idempotent(node):
        node.raw(b"NET DOWN\nNET DOWN\n"); out = node.pump(1.5)
        n = sum(1 for _, l in out if "state=DISCONNECTED" in l)
>       assert n == 1, f"{n} DISCONNECTED transitions for 2 NET DOWN"
E       AssertionError: 2 DISCONNECTED transitions for 2 NET DOWN
E       assert 2 == 1

tests/test_iot_recommendations.py:233: AssertionError
_____________________________________
test_rec6d_net_up_when_connected_is_silent
_____________________________________

node = <test_iot_recommendations.Node object at 0x7fea46717bb0>

    def test_rec6d_net_up_when_connected_is_silent(node):
        out = node.cmd("NET UP", 2)
>       assert not any("network restored" in l for _, l in out), "network restored
logged while connected"
```

```
E       AssertionError: network restored logged while connected
E       assert not True
E        +  where True = any(<generator object
test_rec6d_net_up_when_connected_is_silent.<locals>.<genexpr> at
0x7fea4691ae40>)

tests/test_iot_recommendations.py:238: AssertionError
========================================== short test summary info
==========================================
FAILED tests/
test_iot_recommendations.py::test_rec1a_buffer_never_exceeds_capacity -
AssertionError: STATUSbuffer=12
FAILED tests/
test_iot_recommendations.py::test_rec1b_dropped_counter_and_log - assert 0
== 4
FAILED tests/
test_iot_recommendations.py::test_rec1c_oldest_dropped_newest_kept -
AssertionError: assert [33, 33, 34, 34, 35, 35, ...] == [29, 30, 31, 32, 33, 34, ...]
FAILED tests/test_iot_recommendations.py::test_rec1d_stress_offline_100 -
AssertionError: {'broker': False,'buffer': 100, 'capacity': 8, 'dropped': 0, ...}
FAILED tests/
test_iot_recommendations.py::test_rec2a_each_buffered_message_published
_once - AssertionError:queued [137, 138, 139, 140, 141], published [137, 137,
138, 138, 139, 139, 140, 140, 14...
FAILED tests/test_iot_recommendations.py::test_rec2b_flush_is_fifo -
AssertionError: assert [142, 142, 14...144, 144, ...] == [142, 143, 14...146,
147, ...]
FAILED tests/
test_iot_recommendations.py::test_rec2c_buffer_empty_and_sent_correct_aft
er_flush - AssertionError: {'broker': True, 'buffer': 0, 'capacity': 8, 'dropped':
0, ...}
FAILED tests/test_iot_recommendations.py::test_rec2e_flush_after_overflow -
AssertionError: expected [172, 173, 174, 175, 176, 177, 178, 179], got [176, 176,
177, 177, 178, 178, 1...
FAILED tests/test_iot_recommendations.py::test_rec3a_no_leak_in_stress -
AssertionError: leak [79.92, 80.12, 80.0, 80.0, 80.0] B/message
FAILED tests/
test_iot_recommendations.py::test_rec3b_no_leak_in_buffer_flush_cycles -
AssertionError: heap lost 3200 B in 5 buffer/flush cycles
FAILED tests/test_iot_recommendations.py::test_rec4a_backoff_grows -
AssertionError: backoff decreases: [500, 50, 50, 50, 50, 50, 50, 50, 50, 50]
FAILED tests/test_iot_recommendations.py::test_rec4b_no_reconnect_storm -
AssertionError: 391 retries in 20s
FAILED tests/
test_iot_recommendations.py::test_rec4c_interval_respects_backoff -
AssertionError: gap 0.001s< backoff 50 ms
FAILED tests/
```

test_iot_recommendations.py::test_rec5a_sent_plus_dropped_equals_generated - AssertionError: {'broker': True, 'buffer': 0, 'capacity': 8, 'dropped': 0, ...}
FAILED tests/test_iot_recommendations.py::test_rec6a_stress_invalid_rejected[-5] - AssertionError: no [CLI]ERROR
FAILED tests/test_iot_recommendations.py::test_rec6a_stress_invalid_rejected[1.5] - AssertionError: no [CLI] ERROR
FAILED tests/test_iot_recommendations.py::test_rec6a_stress_invalid_rejected[abc] - AssertionError: no [CLI] ERROR
FAILED tests/test_iot_recommendations.py::test_rec6a_stress_invalid_rejected[10 20] - AssertionError: no [CLI] ERROR
FAILED tests/test_iot_recommendations.py::test_rec6b_stress_out_of_range_reported[0] - AssertionError: silently clamped (assumption: must report)
FAILED tests/test_iot_recommendations.py::test_rec6b_stress_out_of_range_reported[501] - AssertionError: silently clamped (assumption: must report)
FAILED tests/test_iot_recommendations.py::test_rec6c_net_down_idempotent - AssertionError: 2 DISCONNECTED transitions for 2 NET DOWN
FAILED tests/test_iot_recommendations.py::test_rec6d_net_up_when_connected_is_silent - AssertionError: network restored logged while connected
========== 22 failed, 5 passed in 520.01s (0:08:40) ==========
(.venv) (base) Sergio@SERHIIs-MacBook-Pro homework_buggy %